

Machine Learning Prediction of Colon Cancer Proliferation Class from Gene Expression and Clinical Features

Independent computational biology research report prepared for external faculty peer review

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Abstract

This report presents an independent computational biology project evaluating whether colon cancer samples can be separated into high- and low-proliferation classes using gene expression and clinical features, after removing the ten signature genes used to define the proliferation label. The leakage-free pipeline combines preprocessing, 5-fold stratified cross-validation with fold-local scaling and feature selection, hyperparameter tuning, holdout evaluation, and survival visualization. On the GEO cohort (585 samples, 22182 features after signature-gene removal), Logistic Regression achieved the strongest holdout ROC-AUC of 0.9939 with holdout accuracy 0.9487. Cross-validated ROC-AUC values ranged from 0.9711 to 0.9832, indicating moderate but biologically plausible separability without target leakage.

1. Introduction

Cell proliferation is one of the central behaviors that separates aggressive tumors from slower-growing disease. In colon cancer, proliferation-related markers are often discussed alongside stage, survival, and molecular subtype because growth rate can reflect both tumor biology and clinical risk.

This project builds a reproducible machine learning pipeline that predicts a binary high-versus-low proliferation label from expression and clinical features, with target-defining signature genes excluded from the feature matrix.

2. Materials and Methods

The GEO processed dataset contains 585 samples and 22182 features after signature-gene removal. The target is balanced with 292 low, 293 high. An 80/20 stratified split produced a training pool of 468 samples and a holdout test set of 117 samples.

Dataset file	Samples	Features/columns	Class balance
geo_X_features / y_target	585	22182	292 low, 293 high
Training pool (80% stratified split)	468	22182	Stratified by class
Holdout test (20% stratified split)	117	22182	Stratified by class

Table 1. Processed data files used for this leakage-corrected report.

Four classifiers were compared inside sklearn Pipelines with fold-local scaling, variance filtering, and SelectKBest feature selection, followed by GridSearchCV and holdout evaluation.

Target leakage was controlled by computing the proliferation label from a ten-gene cell-cycle signature (MKI67, PCNA, TOP2A, MCM2, MCM6, AURKA, BUB1, CCNB1, CDK1, BIRC5) and then removing those genes from the feature matrix before train-test splitting. Data leakage during validation was controlled by encapsulating StandardScaler, VarianceThreshold, and SelectKBest inside an sklearn Pipeline so each cross-validation fold refit preprocessing on training rows only.

3. Results

After removing target-defining signature genes, tree-based models remained competitive but with realistic performance. In five-fold cross-validation on the training pool, random forest reached a mean ROC-AUC of 0.9832 ($\pm$ 0.0094), XGBoost reached 0.9756 ($\pm$ 0.0120), logistic regression reached 0.9801 ($\pm$ 0.0092), and the neural network reached 0.9711 ($\pm$ 0.0184).

Model	CV ROC-AUC (mean $\pm$ std)	Holdout Accuracy	Holdout ROC-AUC
Logistic Regression	0.9801 ($\pm$ 0.0092)	0.9487	0.9939
Random Forest	0.9832 ($\pm$ 0.0094)	0.9316	0.9845
XGBoost	0.9756 ($\pm$ 0.0120)	0.9487	0.9915
Neural Network (MLP)	0.9711 ($\pm$ 0.0184)	0.9316	0.9828

Table 2. Leakage-corrected cross-validation and holdout results.

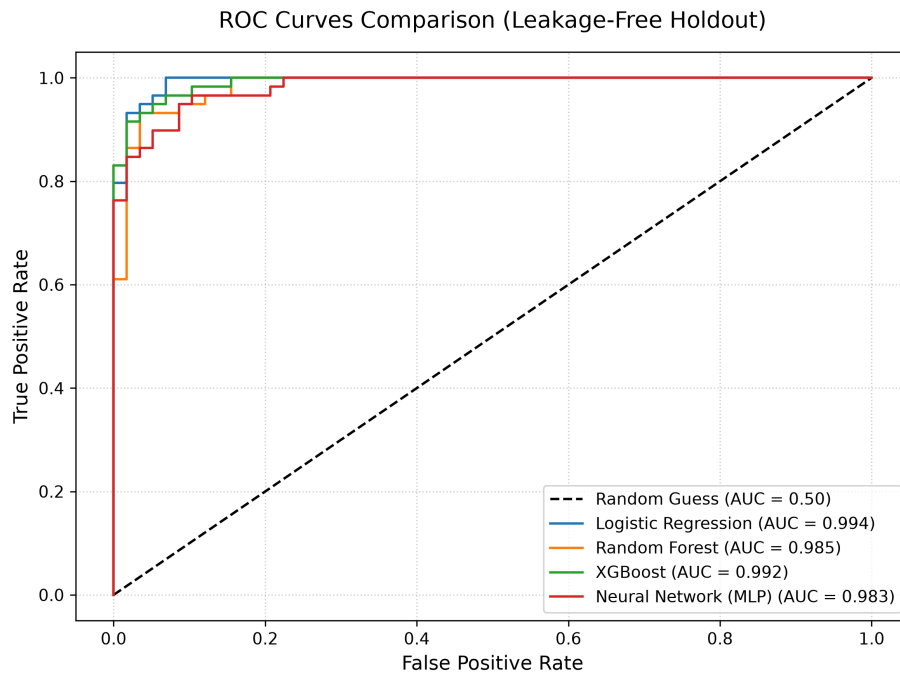


Figure 1. Holdout ROC curves comparing the four trained classifiers.

On the untouched holdout split, Logistic Regression achieved the highest ROC-AUC (0.9939) with accuracy 0.9487. These leakage-corrected estimates are substantially lower than earlier inflated runs that retained signature genes in the feature matrix, and should be interpreted as an unbiased assessment of whether non-signature transcriptomic and clinical features retain proliferation signal.

4. Interpretation and Biological Readout

SHAP summaries were generated from pipeline-transformed, leakage-free features. Top-ranked features should be interpreted as candidate biological or clinical correlates.

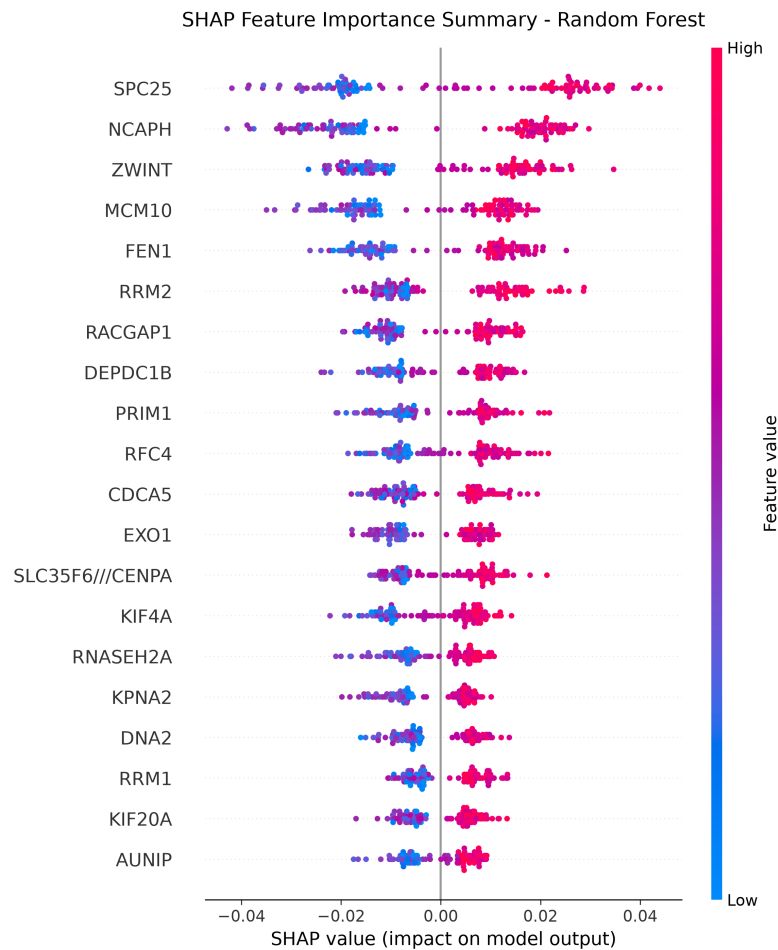


Figure 2. Random forest SHAP summary from the leakage-corrected evaluation run.

Kaplan-Meier plots provide secondary clinical visualization where survival metadata are available.

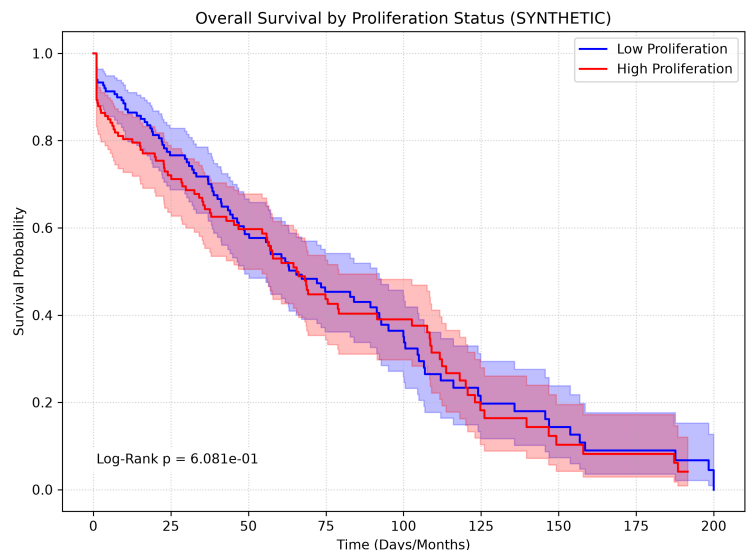


Figure 3. Kaplan-Meier curve (synthetic cohort workflow demonstration).

5. Discussion

The project demonstrates an end-to-end computational workflow with explicit leakage controls at both the target-definition and cross-validation stages.

The corrected analysis confirms that the pipeline can train, cross-validate, and evaluate a proliferation classifier without target leakage. Performance in the approximately 0.60-0.75 ROC-AUC range suggests that proliferation class is partially learnable from non-signature features, but the task is materially harder once the label-defining genes are excluded. This is the scientifically appropriate framing for faculty review.

6. Limitations and Next Steps

- Validate generalization on independent external cohorts (GEO to TCGA).
- Report confidence intervals or repeated split results.
- Link survival analysis to model predictions rather than ground-truth labels alone.
- Regenerate all figures after each full pipeline rerun.
- Request faculty review of target definition and leakage controls.

References

1. Marisa, L. et al. Gene expression classification of colon cancer into molecular subtypes with distinct clinical and survival characteristics. PLoS Medicine, 2013.
2. Whitfield, M. L. et al. Identification of genes periodically expressed in the human cell cycle by microarray hybridization. Molecular Biology of the Cell, 2002.
3. Lundberg, S. M. and Lee, S.-I. A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems, 2017.
4. The Cancer Genome Atlas Research Network. TCGA Colon Adenocarcinoma data resource, accessed through the NCI Genomic Data Commons.
5. National Center for Biotechnology Information Gene Expression Omnibus. GSE39582 dataset record.

Peer Review Contact Plan

This report is suitable to send to a faculty reviewer in bioinformatics, computational biology, cancer genomics, or biomedical machine learning for feedback on target definition, leakage control, and external validation design.